

Alloplastic reconstruction of the temporomandibular joint in treatment of craniofacial developmental or congenital anomalies: a surgical case report.

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To describe the surgical planning and treatment approach in addressing the oral and maxillofacial needs for one particular patient suffering congenital anomalies of the craniofacial anatomy. A secondary objective is to restore mandibular and maxillary function and esthetics to the young, adult patient. The tertiary objective is to educate the surgeon as to alloplastic implant options that may be available to restore function and relieve pain for the patient missing certain anatomical structures from birth. Through the use of advanced medical imaging tools, a thorough understanding of the patient's prior failed medical treatments, and a meticulous understanding of the benefits of alloplastic reconstruction, the surgeon can mitigate patient symptoms and at the same time increase the probability of a successful outcome. The surgeon can then collaborate with the surgical and implant design team to not only prescribe the surgical correction necessary, but also design and construct the actual temporomandibular joint (TMJ) and mandibular implants used for the case, as well as perform mock surgery on a stereolithography anatomical (SLA) model needed to restore function and esthetics for the patient. The end result is that a Patient-Specific (custom) (TMJ Implants, Inc., Golden, CO, USA) prosthesis can then be manufactured and adapted surgically to the patient's unique anatomy. The early results from this alloplastic reconstructive surgery have brought forth the anticipated results of replacement of the congenitally absent TMJ, improvement of jaw function, reduction of joint pain, as well as improvement of the esthetics. Without this breakthrough surgical development, many of these congenital anatomically deficient patients would not be able to enjoy normal breathing, mastication, jaw function, esthetics, oral and dental health, and the emotional relief that these corrections allow. Many patients who suffer birth-related anatomical deficiencies (such as cleft lip and branchial arch syndrome) never receive the surgical correction necessary. Many have had autogenous reconstruction of missing mandibular and TMJ structures only to relapse, at times, into a more disfiguring and lasting condition. By simply placing a Christensen Fossa-Eminence Prosthesis (FEP) and Condylar Prosthesis (CP) in a total joint-replacement solution, one is more likely to achieve satisfactory TMJ mobility and reduction in pain, as well as the mechanical replacement of missing anatomical structures and predictable and lasting improvement. As an additional benefit, the patient also would enjoy enhanced esthetics.

Accuracy of digital templates for guidance of custom-made total temporomandibular joint replacement.

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This study evaluated the application of digital templates to guide custom-made total temporomandibular joint (TMJ) replacement. Patients treated with a custom-made total TMJ prosthesis for TMJ osteoarthritis, ankylosis, or tumours were included prospectively. Before surgery, two types of digital template (articular eminence and condyle neck templates) were designed to guide the bone osteotomy and prosthesis positioning. The reconstructive outcomes were assessed through clinical examinations and accuracy analysis by superimposing the postoperative three-dimensional craniomaxillofacial model onto the preoperative virtual plan. Thirty-seven patients (45 joints) underwent successful TMJ reconstruction with the custom-made TMJ prosthesis guided by the digital templates, without intermaxillary fixation. For all patients, the occlusal relationship was stable in reference to the preoperative state. There were significant improvements in pain and maximum inter-incisal opening. Through merging of the pre- and

postoperative craniomaxillofacial models, the maximum implanted error was 1.17 ± 0.23 mm in linear measurement and 1.19 ± 0.14 mm in surface deviation. There was more deviation in the anteromedial part of the fossa and inferior part of the mandibular handle. The digital templates were able to assist in the accurate placement of the TMJ prosthesis without the need for intermaxillary fixation.

Utilization of Alloplastic Prostheses in the Extended Reconstruction of the Temporomandibular Joint: A Review of the Literature.

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Extended temporomandibular joint reconstruction (eTMJR) can be described as a refined approach to conventional temporomandibular joint reconstruction (TMJR) designed to address both the articulating components of the TMJ and associated mandibular segmental defects or defects in the skull base. Extended temporomandibular joint reconstruction (eTMJR) combined with the advancement in VSP not only offers improved functional and esthetic outcomes but also signifies a significant leap forward in the realm of TMJR interventions. In comparison to autogenous grafts, alloplastic prostheses exhibit superior outcomes concerning MIO, pain management, and dietary functionality, underscoring their potential as the preferred treatment modality. This review article showcases an in-depth exploration of eTMJR, covering its classifications, indications, contraindications, advantages, disadvantages, complications, virtual surgical planning (VSP), criteria for successful alloplastic devices, and the surgical approach.

Total temporomandibular joint prosthesis as a surgical option for severe mouth opening restriction. A case report of a bilateral intervention.

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Several conservative treatment approaches to the disorders of the temporomandibular joint (TMJ) have been described in the literature. Nonetheless, in a minority of cases not responsive to reversible conservative therapies a surgical approach to the TMJ is needed. In recent years, a total temporomandibular joint replacement with alloplastic prosthesis have been introduced as a treatment option in the presence of a severely damaged or mutilated joint, mainly resulting from severe joint diseases, as in the case of complex inflammatory-degenerative diseases, or failure of previous surgeries. The present paper described a case report of a bilateral temporomandibular joint replacement intervention in a female patient with severe mouth opening restriction and pain in the TMJ area. Also, a discussion of the potential indications for TMJ replacement has been provided, along with the description of the surgical procedure.

Prevention and detection of prosthetic temporomandibular joint infections-update.

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Prosthetic joint infections are not only distressing complications for patients and surgeons, but also have an enormous financial impact on healthcare systems. The reported incidence of prosthetic joint infection is likely underestimated due to difficulties in their diagnosis. This unfortunate complication has

challenged joint replacement surgeons for years, despite all the advances made in this surgical discipline. Since eradication of these infections can be very difficult, prevention remains the primary objective. Identifying recipient risk factors, adopting a proper surgical technique, appropriate wound care, optimizing the operating room environment, and appropriate postoperative care have become some of the core elements that can help to minimize the overall incidence of this complication. The purpose of this article is to provide the temporomandibular joint replacement surgeon with an update on the prevention and detection of prosthetic joint infections based on a review of the most recent information published in the orthopedic and surgical literature.

Management of the infected temporomandibular joint total joint prosthesis.

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Patients with specific temporomandibular joint (TMJ) conditions and pathology may benefit from TMJ reconstruction by use of total joint prostheses. A potential risk to patients receiving TMJ total joint prostheses is infection. The purpose of this study was to present our experience in treating infected TMJ total joint prostheses over a 12-year period, as well as the protocol we have developed to manage acute infections and the protocol we follow for chronic infections of TMJ total joint prostheses. This retrospective study evaluated the records of 316 consecutive patients (290 female and 26 male patients) who had TMJ reconstruction from 1997 to 2009 by 1 surgeon, using patient-fitted TMJ Concepts total joint prostheses (TMJ Concepts, Ventura, CA), with a total of 579 prostheses placed, to determine the occurrence and management methods of postoperative infections. Postoperative infections involving the TMJ prostheses developed in 8 of 316 patients (2.5%) and 9 of 579 prostheses (1.6%): 7 patients unilaterally and 1 patient bilaterally. Patients were divided into 2 groups: group 1 (n = 5) had acute infections in 6 joints and group 2 (n = 3) had chronic infections in 3 joints. Patient 5 began in group 1 but was transferred to group 2. One patient with Munchausen syndrome and self-induced infections was eliminated from the study. Patients were treated by our protocol for management of acute or chronic infections. In group 1 (n = 5) the onset of infection symptoms averaged 12 days after surgery (range, 5-24 days). The time from onset of symptoms to surgical intervention was 3.4 days (range, 2-5 days). We found that 4 of 5 patients (80%) and 5 of 6 joints (83%) were successfully treated with retention of the prostheses. Patient 5 varied from the protocol and maintained a chronic infection of her right TMJ prosthesis that transferred her to group 2. In group 2 (n = 3) all 3 patients (including patient 5) had chronic infections with draining fistulas that were successfully treated by the chronic infection protocol including prosthesis removal and replacement. TMJ total joint prostheses can become infected. The earlier the diagnosis is made and the acute infection protocol initiated (within 2-5 days), the greater the chance of salvage of the prosthesis. For chronic infections, the treatment protocol has likewise been very successful, but it does require 2 surgical stages for removal and replacement of the prosthesis. Management of infected total joint prosthesis can be challenging, but with aggressive treatment following the appropriate protocol, infected prostheses can be successfully managed.

Considering total temporomandibular joint replacement.

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Patients with very advanced degenerative disease, ankylosis, post-traumatic condylar destruction, and multi-operated patients may be candidates for joint replacement with fossa and condylar prostheses.

Great advances have been made in developing biocompatible materials, improved designs for patient-fitted prostheses. These devices have treated internal derangement cases after multiple surgical and nonsurgical treatment failures, as well as restoring form and function following the removal of failed Vitek Proplast-Teflon (Houston, Texas) containing temporomandibular joint implants. This paper will provide practitioners dealing with complex, debilitated, functionless temporomandibular joint (TMJ) patients with information related to this treatment modality. They will then be able to address the indications for the use of alloplastic temporomandibular joint replacement devices, the devices presently available, the surgery involved in their placement, possible complications of implantation and post-operative outcomes and expectations with patients who would benefit from the implantation of these devices.

Alloplastic temporomandibular joint replacement - past, present, and future: "Learn from the past, prepare for the future, live in the present." Thomas S. Monson.

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Based on evidence from the orthopaedic, biomedical engineering, and oral and maxillofacial surgical literature, this paper discusses reported successes and failures of past alloplastic temporomandibular joint (TMJ) devices that have led to the development of present total temporomandibular joint replacement (TMJR) devices. The paper concludes with discussion of the ongoing research that will lead to future embodiment (materials, designs, and manufacture) advances in TMJR management of severe and debilitating end-stage TMJ disease, further improving patients' mandibular function, form, and overall quality of life.

Total Joint Replacement for Immediate Reconstruction following Ablative Surgery for Primary Tumors of the Temporo-Mandibular Joint.

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Temporomandibular joint (TMJ) tumors are rare and difficult to diagnose. The purpose of this retrospective study was to evaluate the clinicopathologic characteristics of twenty-one patients with primary TMJ tumors between 2010 and 2019 and to analyze the surgical outcome and morbidity after ablative surgery and TMJ replacement. This case series confirmed the difficulty of diagnosis and reaffirmed the need for early recognition and management of TMJ tumors. There were no pathognomonic findings associated with TMJ tumors, although single or multiple radiopaque or radiolucent areas were observed on plain or panoramic radiographs. Occasionally, bone resorption or mottled densities caused by pathologic calcification and ossification were seen. Computed tomography and magnetic resonance imaging played an important role in the diagnosis. In our study, the distribution of histologic types of TMJ tumors was quite different from that of other joint tumors. The recommended treatment was surgical intervention by ablation of the joint and TMJ replacement. The results of this retrospective study support the surgical exeresis and replacement with TMJ stock and custom-made prostheses and show that the approach is efficacious and safe, reduces pain and improves mandibular movements, with few complications.

Does Metal Hypersensitivity Have Relevance in Patients Undergoing TMJ Prosthetic Replacement?

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Although concerns regarding implant-related metal hypersensitivity exist, no guidelines have been accepted for screening or treatment of patients with metal sensitivity before temporomandibular joint replacement (TMJR). The present study aimed to determine the trends in TMJR metal sensitivity screening and the effect of positive test results on patient treatment. A questionnaire was distributed to members of the American Society of Temporomandibular Joint Surgeons and the European Society of Temporomandibular Joint Surgeons using a web-based data collection and analysis tool. The questions aimed to establish the screening methods used to establish a diagnosis of metal hypersensitivity and any changes in management when metal hypersensitivity had been documented. A total of 72 responses (32.43%) were collected through the survey, with respondents primarily practicing in the United States (47.22%). Of the participants, 75% believed that metal hypersensitivity occurs with TMJR and only 2.77% did not. Most agreed that patients should be questioned about any metal sensitivities before consideration for TMJR (91.66%), and 80.55% routinely asked their patients if they had a previous sensitivity to any metals. Nickel was the most commonly encountered metal allergen (64.28%). If a metal allergy were reported, 44.28% of surgeons would refer the patient to determine the specific metal allergen. The diagnosis for metal allergy was based primarily on patient signs and symptoms (52.85%) and an allergist or dermatologist consultation (44.28%). The lymphocyte transformation test was the test most used by respondents to confirm the diagnosis. Of the surgeons, 41.42% would choose to observe/monitor as their first choice for patients testing positive for metal hypersensitivity. In the case of a "mild" reaction to metal hypersensitivity testing, 54.28% would alter their implant choice to a nonreactive metal component. For "severe" reactions to preoperative metal hypersensitivity testing, 62.85% would alter their implant choice to a nonreactive metal component, and 22.85% would choose not to perform TMJR. The results from the present study have demonstrated that respondent TMJ surgeons agree that metal hypersensitivity occurs with TMJR. However, their approaches to screening and managing metal hypersensitivity vary. Further research of this topic is required to eventually develop specific management pre- and postoperative guidelines for the treatment of patients with metal sensitivities.